

Policy Lab Simulator: AI-Powered Micro-Economy Sandbox

Overview

The **Policy Lab Simulator** is an AI-driven sandbox that models how policy changes affect communities — like schools, small institutions, or student populations. Users can modify virtual 'policies' (e.g., class timings, resource distribution, event frequency) and instantly observe simulated outcomes on agents (students, teachers, administrators) driven by reinforcement learning and adaptive reasoning models.

Instead of relying on static assumptions, the simulator evolves its environment dynamically — letting policymakers *experiment, visualize, and learn* from data-informed virtual societies.

Core Idea

Actors: Each student/teacher/administrator is an AI agent with traits like motivation, fatigue, social influence, and academic performance.

Policies: Global levers like scheduling, incentives, infrastructure, or event frequency.

World Dynamics: Determines how these factors interact — e.g., later start times reduce fatigue but lower after-school productivity.

AI Layer: Agents evolve behaviors through *Reinforcement Learning* (reward = satisfaction, performance, attendance, etc.) and *Meta-Reasoning Models* that simulate complex decision-making beyond numeric optimization.

Tech Stack

Component	Technology
Simulation Engine	Python (SimPy / Mesa)
Agent AI	PyTorch / TensorFlow (for RL)
Reasoning Layer	OpenAI API / Local LLM (for qualitative behavior)
Data & Visualization	Pandas, Plotly/Dash or Streamlit
Backend	FastAPI / Flask
Frontend	Streamlit / React (optional)

6-Week Development Roadmap

Week	Milestone
1	Define environment structure (agents, world variables, policy inputs). Build base simulation loop.
2	Add agent classes with core behaviors (attendance, motivation). Introduce adjustable policies.
3	Integrate reinforcement learning — train agents to optimize personal outcomes.
4	Create dashboards to visualize metrics (attendance, satisfaction, performance trends).
5	Add reasoning/LLM layer for qualitative reactions to new policies.
6	Polish front-end, run scenario demos (e.g., 'What if classes start at 10 AM?'). Prepare presentation visuals.

Why It Matters

The Policy Lab bridges human and algorithmic reasoning — creating an interactive digital twin for institutions. It gives schools and colleges a *safe, data-informed space* to test reforms before implementation, combining simulation, AI, and storytelling into one cohesive tool.